GEK-36495

# INSTRUCTIONS

## ELECTROHYDRAULIC CONTROL

## FOR

## BOILER FEEDPUMP TURBINE

## 3S7513TC300A15

GENERAL ELECTRIC

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*These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company*

## INTRODUCTION

The electronic portion of an electrohydraulic control system for a boiler feedpump turbine drive is described in these instructions.

Any adjustment or maintenance of this equipment should be performed either by qualified turbine personnel or supervised by such personnel.

This equipment is an electronic speed governor whose speed set point can be varied by an automatic boiler controller or by an operator from a remote control unit. Electronic redundancy and mechanical overspeed trip are used to provide a high degree of reliability and safety.

The elementary diagram and alignment diagram for this unit are included in these instructions. Since both the number and arrangement of auxiliary components vary with particular application, it is necessary to refer to the drawings furnished with each equipment to determine the correct connections and auxiliary components furnished in each instance.

## RECEIVING, HANDLING AND STORAGE

### RECEIVING AND HANDLING

Immediately upon receipt, the equipment should be carefully unpacked to avoid damaging the apparatus. Particular care should be exercised to prevent small parts being mislaid or thrown away in the packing material.

As soon as the equipment is unpacked it should be examined for any damage that might have been sustained in transit. If injury or rough handling is evident, a damage claim should be filed immediately with the transportation company and the nearest General Electric Sales Office should be notified promptly.

### STORAGE

If the equipment is not to be used as soon as it is unpacked, it should be stored in a clean, dry place and protected from accidental damage. Particular care should be exercised to avoid storing the equipment in locations where construction work is in progress. If the equipment is to be placed in long term storage, the interior cabinet temperature should be 5°C above ambient.

## DESCRIPTION

### SPEED CONTROL

The speed control loop will be described first. Starting at zone B6 and B7 of the alignment drawing, the coil of the electrohydraulic servo valve receives a low power signal from the electronic unit and controls the flow of oil to a double acting hydraulic cylinder which positions the turbine steam control valve. Turbine speed is sensed by three magnetic sensors mounted near the periphery of a shaft-driven gear. Pulses whose frequency is directly proportional to turbine

## DESCRIPTION (continued)

SPEED CONTROL (cont'd)

speed are generated as each gear tooth passes a sensor. Primary and secondary speed channels in the electronic control produce a DC voltage proportional to the frequency of the sensor outputs. Outputs of two speed channels are fed into high value gate number one, which rejects the lower of the two speed signals and passes on the higher valued speed signal. This redundancy permits one channel to fail without significantly affecting turbine speed. The secondary channel is adjusted for an output 1% less than the primary channel. It is therefore uncontrolling, except in case of failure of the primary channel or if a backup amplifier test switch in the cabinet is actuated which increases the output of the secondary channel to 1% greater than the primary channel. A test meter in the cabinet can be used to determine the controlling channel.

The speed signal from high value gate one is summed with the speed set point to obtain a speed error. (See zone B3 of the alignment diagram.) The speed error is passed through a stability compensation and error amplifier, which includes a regulation adjustment to obtain a valve lift demand signal. This is summed with a valve position signal to obtain a valve position error which is passed through the servo amplifier to the servo valve. Increasing the speed set point increases valve lift demand, valve position error, and valve opening. Speed will increase until the rise matches the set point increases.

The valve position signal is obtained from a Linear Variable Differential Transformer (LVDT) whose core is positioned by the control valve. Three kilohertz excitation for the LVDT is generated in the electronic cabinet. A demodulator in the cabinet produces a DC signal proportional to the AC output of the LVDT, which varies with core position.

A control valve closing bias is applied directly to the servo coil of the control valve in the tripped condition. Before the EHC can reset the feedpump turbine, a valve closing signal from the servo amplifier (2SA) must be present to operate the Close Signal Check (CSC) relay and remove the bias signal that had been previously applied directly to the servo valve coil. Relay circuitry also requires that the electrical connectors must be plugged into the servo valve, the LVDT, and the speed sensors before starting. The LVDT signal is biased to close the control valve if the LVDT connector is removed with the turbine running. The servo valve contains a 5% mechanical bias which will close the control valve in the absence of an electrical signal.

SPEED INDICATION

At zone B9 of the alignment diagram, a third magnetic sensor and circuitry identical to the previously described speed signal are used for speed indication. An analog signal is obtained for speed indication on the EHC remote control panel, local panel, and customer use.

## DESCRIPTION (continued)

SPEED INDICATION (cont'd)

A clamp signal from the 10% speed voltage comparator holds the turbine to less than 10% speed if both the primary and secondary speed channels show less than 10% speed.

At 105% speed, a voltage comparator operates a relay that unlocks the lock out valve in the emergency trip exerciser scheme. At 110% speed another voltage comparator operates the 110SX relay to indicate that 110% speed has been reached. A contact of this relay is provided for the customer.

SPEED REFERENCE OR SET POINT.

Identical motor driven LVDT's are used to generate the manual and automatic speed references. The core position is varied by a two-phase stepping motor which moves in discrete steps and locks in position when not running. Raise and Lower pushbuttons on the operator's manual control panel cause the manual reference to run from the LSS stop to the HSS stop in 80 to 90 seconds. Raise Fast and Lower Fast pushbuttons will run the manual reference from stop to stop in approximately 10 seconds. The automatic speed reference circuits contain interface circuitry for compatibility with a specific boiler controller. The auto speed reference is adjusted to run from the LSS to the HSS in 10 seconds. The automatic and manual speed references are fed into a low value gate which rejects the higher reference and passes on the lower reference to control turbine speed. A relay operated from this gate indicates which of the two references is controlling.

An overspeed test can be performed after first running both manual and automatic references to the high speed stop. The Overspeed Test pushbutton is then actuated, causing an additional speed reference to be generated which increases linearly with time at approximately 12% speed per minute. This is added to both manual and automatic references. If turbine speed rises to 120% of rated, a speed signal will override the ramp generator and prevent further speed increase. Upon release of the Test pushbutton, the reference ramps back to normal at the same rate as it increased.

HIGH PRESSURE/LOW PRESSURE STOP VALVE INDICATION

The EHC cabinet contains circuitry for driving HPSV and LPSV position indicators located on the control panel supplied with the EHC cabinet. The signals for these position indicators are obtained from LVDT's mounted on their respective stop valves. This function is for position indication only and is not part of the control circuitry for the feedpump. (Refer to zone C7, C8 of the alignment drawing.)

## DESCRIPTION (continued)

LOW SPEED SWITCH

An electronic Low Speed Switch (LSS) is provided in the EHC cabinet to automatically, if selected, engage the turning gear on decreasing turbine speed less than 2 rpm.

If at turbine speeds greater than 200 rpm, pulses are lost from the LSPU shown at zone A9 of the alignment drawing, the LSS will prevent the LSSX relay from energizing and prevent automatic engagement of the turning gear by the EHC.

The LSS provides an alarm under the following conditions:

1. If the turbine is at standstill, 0 rpm, and then is increased in speed and pulses are not detected from the LSPU, the LSS 2K contacts will energize the LSS alarm X relay when 200 rpm is reached.

2. If the pick-up pulses are detected from the LSPU but lost before 200 rpm is reached, the LSS 2K contacts will energize the LSS alarm X relay when 200 rpm is reached.

3. If pick-up pulses are lost from the LSPU at turbine speeds greater than 200 rpm, the LSS 2K contacts immediately energize the LSS alarm X relay.

4. Once the LSS alarm X relay is energized, the automatic engagement of the turning gear is prevented until pulses from the LSPU are obtained.

POWER SUPPLIES

Two redundant power supplies, referred to as line A and line B, are provided, They are the full wave rectifier, capacitor filtered variety. Each receives its 120 VAC from the customer's power.

The 120 VAC is fed to a transformer that has three secondary windings. The output of each of the rectified, filtered supplies is connected to three identical series regulators that are adjusted for 30 VDC, 24 VDC and -22 VDC.

The outputs of the series regulators on the line A supply are adjusted slightly higher so they are carrying load and the line B supply is the backup power supply.

SERIES REGULATOR

Operation of the series regulators is described as follows. Output of the voltage sensing bridge consisting of 15R, 16R, 17R, 1P and 2ZD is fed to 7Q and 8Q. An increase in output voltage raises the voltage on 8Q, increasing its conduction and lowering conduction of 7Q. Decreased current in 7Q decreases current in 4Q, 5Q, and 6Q, dropping the output voltage until balance is restored. Excessive current output through 10R turns 3Q on, 2Q off, and 1Q on, grounding the base of 7Q. With 7Q off, series regulators 4Q, 5Q, and 6Q are off and the output goes to zero. When 1C discharges, 3Q is turned off and output is restored. The circuit will oscillate between off and on until the cause of the high current is removed.

## MONITORING

Located in the front of the cabinet are two meter panels that contain several alarms, meters, indicating lights and test meter that will allow checking or monitoring of the power supplies and most amplifiers in the cabinet.

POWER SUPPLY ALARM

For example, an alarm will occur if the 30 volt line A power supply should exceed the 95% or 105% set point of the meter relay in the cabinet. There will be three lights that will be illuminated. First, the power lamp on the control panel supplied with the cabinet will change from a steady glow to a flashing indication, the alarm lamp in the cabinet will be illuminated and the illuminated pushbutton above the line A 30 volt meter will indicate either a high or low voltage.

The three alarm lamps can be cleared after the malfunction is corrected by depressing the lighted pushbutton above the line A 30 volt meter. Identical functions are provided for both the line A and line B 30 V, 24 V, and -22 V power supplies and 3 KHz 6 V RMS power supply in the EHC cabinet. If any of these power supplies are not at their proper voltage level when the 120 V 60 Hz is applied, startup of the feedpump will be prevented.

GROUND ALARM (24 VDC)

If a ground were to occur on the positive or negative side of either the line A or line B 24 volt power supply, the power lamp on the control panel would flash, the +24 V GND/-24 V GND lamp in the cabinet would be illuminated, and the alarm lamp in the cabinet would be illuminated. These lamps will be cleared when the ground is removed. Depressing the +24, -24 V lamp in the cabinet is not required to clear the lamp.

HSS/LSS ALARM

HSS and LSS indication is provided both in the cabinet and on the control panel for both the manual and auto speed reference travel limits. These lamps will be cleared when the respective speed reference is driven off the limit.

CABLE CHECK ALARM

The cable check will prevent startup of the feedpump if the cables are not properly connected. The power lamp on the control panel will flash, the cable check lamp in the cabinet will be illuminated, and the alarm lamp in the cabinet will be illuminated until the cable is properly reconnected.

LAMP TEST

A lamp test can be performed on the indicating lamps in the cabinet by depressing the Alarm pushbutton. There aren't any lamps in the Backup Speed Amplifier Test pushbutton.

## MONITORING (continued)

LAMP TEST (cont'd)

Depressing the Power pushbutton on the control panel will test the lamps on the control panel. There aren't any lamps in the Lower Fast, Lower, Raise, Raise Fast pushbuttons.

TEST SWITCH

The amount of current being supplied by each series regulator can be observed by the test switch and meter in the cabinet. When measuring series regulator current, the meter is calibrated to read approximately 5 amps full scale.

The voltage output of amplifiers 1 thru 12 can be observed by using the test switch and meter. When measuring amplifier outputs, the meter is calibrated to read 10 volts full scale. The amplifier numbers and function are shown on the alignment drawing and elementary diagram.

## INSTALLATION AND ADJUSTMENT

The electronic control cabinet should preferably be mounted in a clean, dry, air-conditioned environment. Ambient temperature should not exceed 104°F unless otherwise noted on the outline drawing for a specific unit.

Equipment adjustment and the required test equipment are covered in detail in an alignment diagram and alignment instructions furnished with each unit.

The location of subassemblies within the cabinet is shown on the elementary diagram furnished with each control. Printed circuit board locations are shown on sheets 1 and 2 of the elementary diagram.

## OPERATION

Refer to the instruction book for the turbine.

## MAINTENANCE

No periodic maintenance is required. Outage time due to component failure is best minimized by maintaining a complete set of spare parts. Elementary diagrams of the printed circuit boards are included in this instruction book. The assembly drawings showing component location on the printed circuit boards are furnished with each order except where component identifications are marked on the boards.

## TROUBLESHOOTING

Most troubleshooting problems can be quickly isolated to a particular section of the control by use of the built-in test meter, the meter relays, and alarm lights. These, used in conjunction with the alignment diagram, indicate the operating condition of each section. An oscilloscope is recommended for pinpointing a defective component or components. Each printed circuit board can be removed from the rack for servicing without disconnection due to the use of terminal boards.

## RENEWAL PARTS

When ordering renewal parts, address the nearest Apparatus Sales Office of General Electric Company and supply the following information.

a. Catalog number stamped on the part and a complete description of the part, including its use and location.

b. Complete nameplate data appearing on the assembly of which the part is a component.

c. If possible, data on original order on which equipment was first supplied, including all numerical references.

Since operation of a steam turbine-generator is dependent upon operation of the feedpump control, it is suggested that an adequate stock of spare parts be carried on hand to minimize outage time due to component failure.

## METER RELAY LAMP REPLACEMENT

Type 195 Meter Relays have been made with two versions of lamps; one without leads attached to the lamp (GE Pt. No. 4868K32P001) and the other with leads attached to the lamp and lamp holder assembly (GE Pt. No. 4872K12G700). These replacement parts can be ordered directly from: the nearest General Electric Sales Office or Utility & Process Automation Products Department, General Electric Company, 40 Federal Street, West Lynn, Mass. 01910.

In the event of lamp failure, the applicable lamp or lamp assembly can be replaced when the control unit is detached from the rear of the indicator set point unit. Then, the lamp-access plate (see figure) must be removed (screws are beneath adhesive labels on some models) to expose the lamp holder which can be pulled out.

PT.NO. 4868K32P001 LAMP

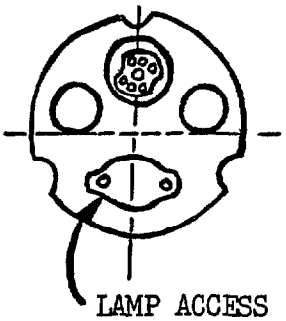
When installing the bayonet-type Pt. No. 4868K32P001 (without leads), twist the lamp in the holder assembly until the lamp filament will be in the horizontal position when the lamp holder is replaced in the indicator set point unit.

PT. NO. 4872K12G700 LAMP & HOLDER

The Pt. No. 4872K12G700 lamp has been fixed in the lamp holder at the factory so the filament position will be in the correct position when the keyed assembly is inserted in the indicator set point unit. Make sure the lamp-lead clips are engaged by the screws when the lamp access cover is replaced.

NOTE: If relay operation is not within $\pm$ 2% of set point index, the lamp holder position must be rotated 180°.

## METER RELAY LAMP REPLACEMENT (continued)




REASSEMBLY

After the new lamp or lamp assembly has been installed, make sure the lamp access plate is replaced (and screws are tight) before control unit is reassembled on rear of indicator set point unit.

# DRAWING LIST

| DESCRIPTION | DRAWING NUMBER |
|---|---|
| Diode Board, 1DB | 44A300243 |
| Reference Low Value Gate, 2LVG | 44A300281 |
| Compensation and Regulation, 1CR | 44A301400 |
| Overspeed Integrator, 1OSI | 44A302692 |
| Signal Low Value Gate, 1LVG | 44A304763 |
| Alignment Instructions | 44A308804 |
| Series Regulator Board #1, 1REC | 44A306392 |
| Renewal Parts | 44A335632 |
| Cable Wiring Diagram, Signal & Low Voltage, 1CO | 44B300423 |
| Control Panel Outline | 44B307064 |
| Low Speed Switch Power Supply, LPS | 44B306144 |
| Cable Wiring Diagram Power, 2CO & 3CO | 44B306148 |
| Series Regulator Board #2, 1SR | 44B306197 |
| Detector Indicator, 1DI | 44C300609 |
| Operational Amplifier, 1OA | 44C300610 |
| Analog Output, 1AO2 | 44C301216 |
| Analog Output, 3AO2 | 44C301622 |
| Frequency to Voltage Converter, 2FVC | 44C301682 |
| Clock and Stepping Motor Drive, CI&SMD | 44C303567 |
| Servo Amplifier, 2SA | 44C304444 |
| 3 KHz Oscillator, 60CS | 44C304462 |
| Signal Converter, 3SC | 44C304898 |
| Voltage Comparator, 1VC | 44C305259 |
| Elementary Drawing, Master | 44C308346 |
| Interconnection Drawing | 44C308347 |
| Low Speed Switch, LSS | 44C306444 |
| Boiler Control Interface, 7BCI | 44C306492 |
| Line A & Line B Power Supply | 44C306526 |
| Cabinet Outline | 44D300043 |
| Alignment Drawing | 44D308104 |
| Control Panel Connection Diagram | 44D308138 |
| Relay Panel Connection Diagram | 44D308137 |
| Summary | 44X308910 |

Drawings Taken from BEK-36495

for BEK-36506

44A300243
44A300281
44A301400
44A302692
44A304763
44A306392
44B300423
44B306144
44B306148
44B306197
44C300609
44C300610
44C301216
44C301622
44C301682
44C303567
44C304444
44C304462
44C304898
44C305259
44C306444
44C306492
44C30652C
44D300043

for BEK-36544

44A308804

1/27/76

3S7931EA184G9 STATIC EXC REG ASM

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 2 | 3S7930NA145G6 | REG PANEL |
| 2 | 3S7932CD160G3 | CONTROL PANEL |
| 2 | 3S7932HA129G9 | MTR OPR POT DRIVE |
| 2 | 44A30384 0-G01 | RECT AND BUS ASM |
| 2 | 44A318979-G01 | PLEXIGLAS |
| 2 | 3S7932MD125G5 | SHAFT VOLT SUPPR |
| 2 | 3S7932MD130G9 | RESISTOR PANEL |
| 2 | 44C372098-G01 | BRKR + RECP ASM |
| 4 | 44A315910-001 | FLD. CIR. BRKR. |
| 4 | 44A339883-001 | FLD CKT BKH HSNG |
| 6 | 44B33476 0-G01 | LINEAR REACTOR |
| 2 | 9RV6A228 | THYRITE |
| 2 | 9145192811 6G4 | RES |
| 8 | 44A37 0867-001 | RECEPTACLE |
| 8 | 100A | LAMP .87A 115V |
| 2 | 44A21844 0-001 | TEST BLOCK |
| 2 | 3S7932YA122A4 | MACH FLD GND DET |
| 4 | 44B232212-011 | THERMOSTAT |
| 36 | 44A33425 0-003 | RECT 250A 600V |
| 36 | 44A33425 0-103 | RECT 250A 600V |

3S7930NA145G6 REG PANEL

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 2 | K8622213F18 | RES 1.8KOHM 10 W |
| 2 | 44A333452-E80 | RES 800 OHM 50 W |
| 2 | 23F1059G202 | CAP, 8 U 600 VDC |
| 6 | 23F1060G202 | CAP, 10 U 600 VDC |
| 4 | 23F1063G202 | CAP, 20U 600VDC |
| 2 | M9729351G10 | POT 10K 50 W |
| 4 | M9729350E25 | POT 250 25 W |
| 4 | 571A135P2 | SUPT |
| 4 | 44B21n645-001 | REACTOR .8H .375A |
| 2 | 395B584G1 | REACTOR 20H .025A |
| 2 | 233B994G1 | REACTOR 7.63MH 5A |
| 4 | 44B21990 7-G01 | REACTOR |
| 2 | 44B211558-G01 | TRANS |
| 2 | 3S7930NA145G1 | RECT ASM |
| 4 | 44B212741-009 | SCR 35A 500V |
| 8 | 1N1206A | RECT 12A 600V |
| 2 | 44B212755-003 | ZENER 50V 3.25W |
| 2 | 44A211331-001 | HEAT SINK |
| 4 | M9729485P1 | SWITCH |
| 2 | 44A333452-E24 | RES 240 OHM 50 W |
| 2 | M9729351F50 | POT 5K OHM 50 W |
| 6 | 44B3124 03-J10 | CAP 1UF 600VDC |

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3S7932CD160G3 CONTROL PANEL

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 4 | TEB122Y100 | CIRCUIT BREAKER |
| 16 | K97747 00P6 | FUSE 20 A 250 V |

| | | |
|---|---|---|
| 2 | K9774700P5 | FUSE 15 A 250 V |
| 4 | K9774701P3 | FUSE 6A 600 V |
| 8 | K9774700P9 | FUSE 35 A 250 V |
| 36 | CR7500F100A | ASM |
| 24 | CR7500F100B | ASM |
| 2 | 4JA5512DH1AD1 | RECTIFIER |
| 2 | IC2800-1622M3H | CONTACTOR |
| 2 | 44A332146-U01 | RELAY |
| 2 | 3GTR39L1 | RELAY |
| 4 | 44A336787-D20 | TIME DELAY RELAY |
| 2 | 3GTR1A35 | ASM |
| 2 | 44A333454-E20 | RES 200 OHMS 225W |
| 2 | 44A333458-E47 | RES 470 OHM 225 W |
| 2 | 44B336196-G01 | FUSE COVER ASM |

3GTR39L1 RELAY

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 2 | P433818363 | ASM |
| 2 | K241406SP1 | SPRING |
| 2 | F3040007P1 | COIL |
| 2 | 128A4080F10F10 | RESISTOR ASM |
| 2 | L2850887G1 | CASE ASM |

P433818363 ASM

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 4 | K2843289G1 | ROD + HANDLE |
| 4 | L2833113G6 | CONTACT ASM |
| 4 | F9454 0 | HOLDING COIL |
| 6 | V478573P2 | SCREW |
| 2 | V24593 01P2 | SCREW |
| 2 | P3816078P11 | BASE |
| 2 | L2850847G2 | DASH POT ASM |
| 2 | M4912246G4 | CORE |
| 2 | 289A503P1 | LEVER |
| 2 | M1687888G6 | CONNECTION |
| 4 | V446086P1 | PIVOT BEARING |
| 2 | V2452440P1 | PIVOT |
| 2 | V1455640P1 | PIVOT BRACKET |
| 4 | V1419522P1 | ARMATURE |

L283084762 DASH POT ASM

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 2 | K8076121G1 | VALVE ASM |

3GTR1A35 ASM

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 2 | P2153698G2 | ASM |
| 2 | L2850887G1 | CASE ASM |
| 2 | F3040073P1 | COIL ASM |
| 2 | 44A318920-E56 | RESISTOR |

REVISE

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P213369862 ASM

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 2 | P2805679P4 | BASE |
| 2 | V8024433G1 | CORE ASM |
| 2 | V400127P1 | PIVOT |
| 4 | V446086P1 | PIVOT BEARING |
| 4 | M1687888G11 | CONN ASM |
| 4 | V1432733P1 | ARMATURE |
| 4 | L2830867G3 | CONTACT SPRING |
| 2 | K2411113P1 | SPRING |
| 8 | V478573P2 | SCREW |
| 4 | K2843289G1 | ROD + HANDLE |

44A318920-E56 RESISTOR

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 2 | 44A333458-E56 | RES 560 OHM 225 W |

3S7932HA129G9 MTR OPR POT DRIVE

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 2 | 3S7932HA127G2 | POT DRIVE ASM |
| 2 | 44B218508-004 | RHEO, ASM, |
| 2 | 44B332086-G01 | COUPLING |
| 4 | 44A318549-005 | RELAY |
| 2 | 44A333460-E51 | RES 510 OHMS 160 W |
| 2 | K9774721L10 | CAP, 250 V A |

3S7932HA127G2 POT DRIVE ASM

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 2 | 44C335182-G01 | PLATE ASM |
| 2 | 44C335183-G01 | PLATE ASM |
| 2 | 44A335708-G01 | PLATE |
| 2 | 44B335094-001 | SHAFT |
| 2 | 44B335094-002 | SHAFT |
| 2 | 44B335094-003 | SHAFT |
| 2 | 44A334391-002 | HARMON DR MOTOR |
| 4 | 44A335700-001 | GEAR |
| 2 | 44A335701-001 | GEAR |
| 2 | 44A335702-001 | GEAR |
| 2 | 44A335812-001 | SPRING |
| 16 | 44A332122-001 | SWITCH MICRO |
| 2 | K5908303G1 | T B ASM |

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44C335182-G01 PLATE ASM

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 4 | 44A335834-005 | BEARING |

44C335183-G01 PLATE ASM

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 2 | 44A335834-005 | BEARING |

| | | |
|---|---|---|
| 2 | 44A335834-007 | BEARING |

44A335708-G01 PLATE

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 2 | 44A335834-007 | BEARING |

44A334391-002 HARMON DR MOTOR

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 2 | 44A332129-G02 | HARM MOTOR DR ASM |

44A332129-G02 HARM MOTOR DR ASM

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 2 | 44A339890-001 | MOTOR |
| 2 | 44A337905-001 | GEARHEAD |
| 2 | N2111P8 | STEEL BALL |

44A335700-001 GEAR

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 4 | 44A335835-024 | SPUR GEAR |

44A335701-001 GEAR

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 2 | 44A335833-096 | SPUR GEAR |

44A335702-001 GEAR

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 2 | 44A335835-120 | SPUR GEAR |

44A308400-G01 RECT AND BUS ASM

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 2 | 44A208094-G07 | FRONT BRIDGE ASM |
| 2 | 44A208094-G08 | REAR BRIDGE ASM |

44A208094-G07 FRONT BRIDGE ASM

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 6 | 44A314200-G01 | RECT, PHASE ASM |
| 2 | 44A316754-001 | KNIFE SWITCH |

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44A314200-G01 RECT, PHASE ASM

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 36 | 44B512352-G01 | POTTED ASM |

44A208094-G08 REAR BRIDGE ASM

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 6 | 44A314200-G02 | RECT PHASE ASM |

| | | |
|---|---|---|
| 2 | 44A316754-001 | KNIFE SWITCH |

44A314200-G02 RECT PHASE ASM

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 36 | 44B312352-G01 | POTTED ASM |

44A318979-G01 PLEXIGLAS

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 24 | K590830G1 | T B ASM |

3S7932MD125G5 SHAFT VOLT SUPPR

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 4 | 23F1064G202 | CAP, 25U 600VDC |
| 2 | IC9033E5D96 | RES 96 OHMS 75W |
| 2 | K590830G1 | T B ASM |

3S7932MD130G9 RESISTOR PANEL

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 6 | IC9033A4D26 | RES 26 OHM 600 W |
| 12 | IC9033B4D13 | RES 13 OHM 565 W |
| 4 | IC9033B4C67 | RES 6,7 OHM 765 W |
| 12 | 571A187P1 | BRKT |

44C372098-G01 BRKR & RECP ASM

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 2 | 44A372252-001 | CIRCUIT BREAKER |

3S7932YA122A4 MACH FLD GND DET

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 2 | 44B334833-G01 | TRANSFORMER |
| 2 | 44A373338-001 | TIME DELAY RELAY |
| 2 | 12PJV11A39 | RELAY |
| 2 | 12HFA54C249 | RELAY |
| 2 | 44A333451-F12 | RES 1200 OHMS 25W |
| 2 | 44A333451-E50 | RES 500 OHMS 25W |
| 2 | 44A337744-G01 | LED POTTED ASM |
| 2 | 44A334235-001 | SWITCH |
| 2 | 44A336210-G01 | TERM BD ASM |

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44A336210-G01 TERM BD ASM

| QTY | PART NUMBER | DESCRIPTION |
|---|---|---|
| 2 | 44A336210-002 | COMP BOARD |
| 12 | 44B232019-010 | RECT 1 A 1000 V |

GENERAL ELECTRIC CO.
WAYNESBORO, VA.

**REGULATOR SUMMARY** 44X308910

JUN 10 1975

**COMPILED BY:** RJ Culver R.J. Culver **DATE:** 6/9/75 1 **SHEETS; SHEET** 1

**CUSTOMER:** LOUISVILLE GAS AND ELECTRIC, MILL CREEK #3

BFTB 122948 Requisition 091-41405

| ITEM | MATERIAL | | QUAN. |
|---|---|---|---|
| 1 | 3S7513TC300A15 Electronic Boiler Feedpump Turbine Control | | 1 |
| | Outline Drawing | 44D300043 | |
| | Wire Table | 44A307299 | |
| | Interconnection Drawing | 44C308347 | |
| | Elementary Drawing | 44C308346 | |
| | Alignment Drawing | 44D308104 | |
| | Alignment Instructions | 44A308804 | |
| | Renewal Parts | 44A335622 | |
| | | | |
| 2 | 3S7513KL300A14 | | |
| | Outline | 44B307066 | |
| | Connection Diagram | 44D305962 | |
| | | | |
| 3 | Cabinet-Control Panel Power Cable | 44B306148 | 2 |
| | | | |
| 4 | Cabinet-Control Panel Control Cable | 44B300423 | 1 |

REVISIONS

**DISTRIBUTION**

| | |
|---|---|
| 11 | |
| FG-W | |
| 12P-FS2 | |
| | |
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| | |

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Diode Board, 1DB. . . . . 44A300243
Reference Low Value Gate, 2LVG . . . . . 44A300281
Compensation and Regulation, 1CR . . . . . 44A301400
Overspeed Integrator, 1OSI. . . . . 44A302692
Signal Low Value Gate, 1LVG . . . . . 44A304763
Alignment Instructions. . . . . 44A308804
Series Regulator Board #1, 1REC . . . . . 44A306392
Renewal Parts . . . . . 44A335632
Cable Wiring Diagram, Signal & Low Voltage, 1CO . . 44B300423
Control Panel Outline . . . . . 44B307064
Low Speed Switch Power Supply, LPS. . . . . 44B306144
Cable Wiring Diagram Power, 2CO & 3CO . . . . . 44B306148
Series Regulator Board #2, 1SR. . . . . 44B306197
Detector Indicator, 1DI . . . . . 44C300609
Operational Amplifier, 1OA. . . . . 44C300610
Analog Output, 1AO2.. . . . . 44C301216
Analog Output, 3AO2 . . . . . 44C301622
Frequency to Voltage Converter, 2FVC. . . . . 44C301682
Clock and Stepping Motor Drive, CI&SMD. . . . . 44C303567
Servo Amplifier, 2SA. . . . . 44C304444
3 KHz Oscillator, 60CS. . . . . 44C304462
Signal Converter, 3SC . . . . . 44C304898
Voltage Comparator, 1VC . . . . . 44C305259
Elementary Drawing, Master. . . . . 44C308346
Interconnection Drawing . . . . . 44C308347
Low Speed Switch, LSS . . . . . 44C306444
Boiler Control Interface, 7BCI. . . . . 44C306492
Line A & Line B Power Supply. . . . . 44C306526
Cabinet Outline . . . . . 44D300043
Alignment Drawing . . . . . 44D308104
Control Panel Connection Diagram. . . . . 44D308138
Relay Panel Connection Diagram. . . . . 44D308137
Summary . . . . . 44X308910

| | |
|---|---|
| Diode Board, 1DB. . . . . | 44A300243 |
| Reference Low Value Gate, 2LVG . . . . . | 44A300281 |
| Compensation and Regulation, 1CR . . . . . | 44A301400 |
| Overspeed Integrator, 1OSI. . . . . | 44A302692 |
| Signal Low Value Gate, 1LVG . . . . . | 44A304763 |
| Alignment Instructions. . . . . | 44A308804 |
| Series Regulator Board #1, 1REC . . . . . | 44A306392 |
| Renewal Parts . . . . . | 44A335632 |
| Cable Wiring Diagram, Signal & Low Voltage, 1CO . . | 44B300423 |
| Control Panel Outline . . . . . | 44B307064 |
| Low Speed Switch Power Supply, LPS. . . . . | 44B306144 |
| Cable Wiring Diagram Power, 2CO & 3CO . . . . . | 44B306148 |
| Series Regulator Board #2, 1SR. . . . . | 44B306197 |
| Detector Indicator, 1DI . . . . . | 44C300609 |
| Operational Amplifier, 1OA. . . . . | 44C300610 |
| Analog Output, 1AO2. . . . . | 44C301216 |
| Analog Output, 3AO2 . . . . . | 44C301622 |
| Frequency to Voltage Converter, 2FVC. . . . . | 44C301682 |
| Clock and Stepping Motor Drive, CI&SMD. . . . . | 44C303567 |
| Servo Amplifier, 2SA. . . . . | 44C304444 |
| 3 KHz Oscillator, 60CS. . . . . | 44C304462 |
| Signal Converter, 3SC . . . . . | 44C304898 |
| Voltage Comparator, 1VC . . . . . | 44C305259 |
| Elementary Drawing, Master. . . . . | 44C308346 |
| Interconnection Drawing . . . . . | 44C308347 |
| Low Speed Switch, LSS . . . . . | 44C306444 |
| Boiler Control Interface, 7BCI. . . . . | 44C306492 |
| Line A & Line B Power Supply. . . . . | 44C306526 |
| Cabinet Outline . . . . . | 44D300043 |
| Alignment Drawing . . . . . | 44D308104 |
| Control Panel Connection Diagram. . . . . | 44D308138 |
| Relay Panel Connection Diagram. . . . . | 44D308137 |
| Summary . . . . . | 44X308910 |